

Ellipses

ANSWER KEY



Standard form

- 1 For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, find:
- a the vertices $(\pm 5, 0)$
 - b the foci $(\pm 4, 0)$
 - c the endpoints of the minor axis $(0, \pm 3)$
 - d the eccentricity $e = \frac{4}{5}$
- 2 Write the standard-form equation of the ellipse with foci $(0, \pm 3)$ and vertices $(0, \pm 5)$.
Major axis vertical with $a = 5$, $c = 3$, so $b^2 = 25 - 9 = 16$: $\frac{x^2}{16} + \frac{y^2}{25} = 1$.
- 3 Consider the ellipse $9x^2 + 4y^2 = 36$.
- a Write the equation in standard form. $\frac{x^2}{4} + \frac{y^2}{9} = 1$.
 - b State the lengths of the major and minor axes. Major axis $2a = 6$ (vertical); minor axis $2b = 4$.
 - c Find the coordinates of the foci. $c^2 = 9 - 4 = 5$, so foci $(0, \pm \sqrt{5})$.

Applications

- 4 A whispering gallery has an elliptical ceiling 40 m long and 24 m high at the center (so $a = 20$ and $b = 12$, with the major axis horizontal). Two people stand at the foci. How far apart are they, to one decimal place?
 $c = \sqrt{400 - 144} = \sqrt{256} = 16$, so the foci are $2c = 32.0$ m apart.
- 5 An ellipse centered at the origin has a horizontal major axis of length 12 and passes through the point $(3, 2\sqrt{3})$. Find its equation.
 $a = 6$: $\frac{9}{36} + \frac{12}{b^2} = 1$ gives $\frac{12}{b^2} = \frac{3}{4}$, so $b^2 = 16$: $\frac{x^2}{36} + \frac{y^2}{16} = 1$.

Ellipses centered away from the origin

- 6 For the ellipse $\frac{(x-2)^2}{16} + \frac{(y+1)^2}{9} = 1$, find:
- a the center $(2, -1)$
 - b the vertices $(2 \pm 4, -1) = (6, -1), (-2, -1)$
 - c the foci $c = \sqrt{16 - 9} = \sqrt{7}$: $(2 \pm \sqrt{7}, -1)$
- 7 Complete the square to write $4x^2 + y^2 - 8x + 6y + 9 = 0$ in standard form, and state the center.
 $4(x^2 - 2x) + (y^2 + 6y) = -9$: $4(x-1)^2 - 4 + (y+3)^2 - 9 = -9$, so $4(x-1)^2 + (y+3)^2 = 4$, i.e. $(x-1)^2 + \frac{(y+3)^2}{4} = 1$. Center: $(1, -3)$.

Eccentricity and orbital applications

- 8 Compute the eccentricity of each ellipse, and state which is more circular (closer to $e = 0$):

a $\frac{x^2}{100} + \frac{y^2}{36} = 1$ $c = \sqrt{100 - 36} = 8$; $e = \frac{8}{10} = 0.8$ **b** $\frac{x^2}{25} + \frac{y^2}{24} = 1$ $c = \sqrt{25 - 24} = 1$; $e = \frac{1}{5} = 0.2$ — this one is more circular

- 9 A planet's elliptical orbit has semi-major axis $a = 150$ million km and eccentricity $e = 0.02$. Find the distance from the center to a focus, c , and hence the closest approach distance to the sun ($a - c$), to two decimal places (in millions of km).

$c = ea = 0.02(150) = 3$ million km. Closest approach $= a - c = 150 - 3 = 147.00$ million km.

Graphing an ellipse

- 10 The upper half of the ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$ is shown (as $y = 2\sqrt{1 - x^2/16}$). Use it to estimate the y -value at $x = 2$, then verify algebraically.

From the graph, $y \approx 1.7$ at $x = 2$. Algebraically: $y = 2\sqrt{1 - 4/16} = 2\sqrt{0.75} \approx 1.73$ ✓.

The ellipse's defining property

- 11 State the geometric (sum-of-distances) definition of an ellipse, and use it to verify that the point $(0, 3)$ lies on the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ with foci $(\pm\sqrt{7}, 0)$.

An ellipse is the set of points where the sum of distances to the two foci is constant ($= 2a$). At $(0, 3)$: distance to each focus $= \sqrt{7 + 9} = 4$, so the sum is $8 = 2a = 2(4)$ ✓, confirming the point lies on the ellipse.

Writing equations from a description

- 12 Write the standard-form equation of an ellipse centered at $(3, 2)$ with a horizontal major axis of length 10 and minor axis of length 6.

$a = 5, b = 3$: $\frac{(x-3)^2}{25} + \frac{(y-2)^2}{9} = 1$.

- 13 Write the standard-form equation of an ellipse with vertical major axis, center $(-1, 4)$, $a = 7$, and $b = 2$.

$\frac{(x+1)^2}{4} + \frac{(y-4)^2}{49} = 1$.

Systems with ellipses

- 14 Find the points of intersection of $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and the line $y = x$.

Substitute $y = x$: $\frac{x^2}{9} + \frac{x^2}{4} = 1 \Rightarrow \frac{4x^2 + 9x^2}{36} = 1 \Rightarrow 13x^2 = 36 \Rightarrow x = \pm \frac{6}{\sqrt{13}}$. Points: $(\pm \frac{6}{\sqrt{13}}, \pm \frac{6}{\sqrt{13}})$ (same sign pairs).

Area of an ellipse

- 15** State the formula for the area of an ellipse with semi-axes a and b , and use it to find the area of $\frac{x^2}{25} + \frac{y^2}{16} = 1$.

Area = πab . Here $a = 5$, $b = 4$: Area = $20\pi \approx 62.83$ square units.

- 16** An elliptical track for a running field has a semi-major axis of 60 m and semi-minor axis of 40 m. Find the exact area enclosed by the track, and give a decimal approximation to the nearest square metre.

Area = $\pi(60)(40) = 2400\pi \approx 7539.8 \text{ m}^2$.

Comparing ellipses of the same eccentricity

- 17** Two ellipses both have eccentricity $e = 0.6$: one has $a = 10$, the other has $a = 20$. Find b for each, and explain why they are the same shape but different sizes (similar ellipses).

For $a = 10$: $c = ea = 6$, $b^2 = 100 - 36 = 64 \Rightarrow b = 8$. For $a = 20$: $c = 12$, $b^2 = 400 - 144 = 256 \Rightarrow b = 16$. Since $\frac{b}{a} = \frac{8}{10} = \frac{16}{20} = 0.8$ is the same ratio for both, the ellipses are geometrically similar (same shape, scaled by a factor of 2).